



BMATS201

USN

Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Mathematics – II for CSE Stream

Time: 3 hrs.

Max. Marks: 100

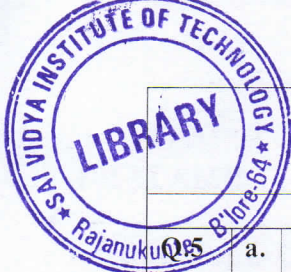
Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. *M*: Marks, *L*: Bloom's level, *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$.	7	L2	CO1
	b.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing into polar coordinates.	7	L3	CO1
	c.	Show that $\beta(m, n) = \frac{\gamma(m)\gamma(n)}{\gamma(m+n)}$	6	L2	CO1
OR					
Q.2	a.	Evaluate $\int_0^1 \int_y^{\sqrt{y}} (x^2 y + xy^2) dx dy$ by changing the order of integration.	7	L3	CO1
	b.	Show that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$	7	L2	CO1
	c.	Using mathematical tools, write the code to find the area of an ellipse by double integration $A = 4 \int_0^a \int_0^{\frac{b}{a}\sqrt{a^2-x^2}} dy dx$, taking $a = 4$, $b = 6$.	6	L3	CO5
Module – 2					
Q.3	a.	Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ along vector $2\hat{i} - 3\hat{j} + 6\hat{k}$.	7	L2	CO2
	b.	Show that the vector $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.	7	L2	CO2
	c.	Prove that the spherical coordinate system is orthogonal:	6	L3	CO2
OR					
Q.4	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z^2 + y^2 - x = 3$ at $(2, -1, 2)$.	7	L2	CO2
	b.	Express the vector $\vec{A} = z\hat{i} - 2x\hat{j} + y\hat{k}$ in cylindrical coordinates.	7	L2	CO2
	c.	Using mathematical tools, write the code to find the curl of $\vec{F} = x^2yz\hat{i} + y^2zx\hat{j} + z^2xy\hat{k}$.	6	L3	CO5

1 of 3



Module – 3

Q.5	a.	Prove that the subset $W = \{(x, y, z) : ax + by + cz = 0; x, y, z \in \mathbb{R}\}$ of the vector space $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$.	7	L2	CO3
	b.	Determine the following vectors are linearly independent or not, $x_1 = (2, 2, 1)$, $x_2 = (1, 3, 7)$ and $x_3 = (1, 2, 2)$ in $\mathbb{R}^3$.	7	L2	CO3
	c.	Show that the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $T(x, y) = (x + y, x - y, y)$ is a linear transformation.	6	L2	CO3

OR

Q.6	a.	Determine whether the vectors $v_1 = (1, 2, 3)$, $v_2 = (3, 1, 7)$ and $v_3 = (2, 5, 8)$ are linearly dependent or linearly independent.	7	L2	CO3
	b.	Verify the Rank-Nullity theorem for the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.	7	L2	CO3
	c.	Consider the vectors $u = (1, 2, 4)$, $v = (2, -3, 5)$, $w = (4, 2, -3)$ in $\mathbb{R}^3$. Find: i) $\langle u, v \rangle$ ii) $\langle u, w \rangle$ iii) $\langle v, w \rangle$ iv) $\langle (u + v), w \rangle$	6	L2	CO3

Module – 4

Q.7	a.	Find an approximate value of the root of the equation $x^3 - x^2 - 1 = 0$, using the Regula-Falsi method upto four decimal places of accuracy, where root lies between 1.4 and 1.5.	7	L2	CO4										
	b.	Using Newton's divided difference formula evaluate $f(4)$ from the following: <table border="1"><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	7	L2	CO4
x	0	2	3	6											
f(x)	-4	2	14	158											
	c.	Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ by using Trapezoidal rule by taking 7 ordinates.	6	L3	CO4										

OR

Q.8	a.	Find an approximate root of the equation $x \log_{10} x - 1.2 = 0$ corrected to five decimal places where root lies near 2.5 by Newton-Raphson method.	7	L2	CO4												
	b.	The area A of a circle of diameter d is given for the following values. Calculate the area of a circle of diameter 82 by using Newton's forward interpolation formula. <table border="1" data-bbox="509 1538 976 1610"> <tr> <td>d</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr> <tr> <td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr> </table>	d	80	85	90	95	100	A	5026	5674	6362	7088	7854	7	L2	CO4
d	80	85	90	95	100												
A	5026	5674	6362	7088	7854												
	c.	Use Simpson's $1/3^{\text{rd}}$ rule to find $\int_0^{0.6} e^{-x^2} dx$ by taking seven ordinates.	6	L2	CO4												

Module – 5

Q.9	a.	Find by Taylor's series method the value of y at x = 0.1 to five places of decimals from $\frac{dy}{dx} = x^2y - 1$ with an initial condition $y(0) = 1$.	7	L2	CO4
	b.	Using the Runge-Kutta method of fourth order solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at x = 0.2 taking h = 0.2.	7	L2	CO4
	c.	Given that $\frac{dy}{dx} = x^2(1 + y)$ and $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$ and $y(1.3) = 1.979$. Compute y at x = 1.4 by applying Milne's method.	6	L2	CO4
OR					
Q.10	a.	Using modified Euler's method, solve $\frac{dy}{dx} = 3x + \frac{y}{2}$ at x = 0.1 corrected to four decimal places by taking h = 0.1, with initial condition $y(0) = 1$.	7	L2	CO4
	b.	Given that $\frac{dy}{dx} = x - y^2$ and $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$. Compute $y(0.8)$ by Milne's method.	7	L2	CO4
	c.	Using mathematical tools, write the code to find the solution of $\frac{dy}{dx} = 1 + \frac{y}{x}$ at y(2) taking h = 0.2. Given that $y(1) = 2$ by Runge-Kutta method of 4 th order.	6	L3	CO5

